Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 6: Velocity–Time Graphs — ANSWERS

One graph runs through Sections 1 to 3. The tram below pulls away from a stop, holds a steady speed, then brakes for the next one. Read the gradient for acceleration, read the area for distance, then use that distance for average speed — the same graph, asked three different ways. Section 4 is a new graph, and this time you draw it yourself.

You met the gradient formula on *Kinematics 5*, on a position–time graph, where it gave you a velocity. It is the same formula here. The only thing that changes is what is up the side of the graph — and that changes what the answer means.

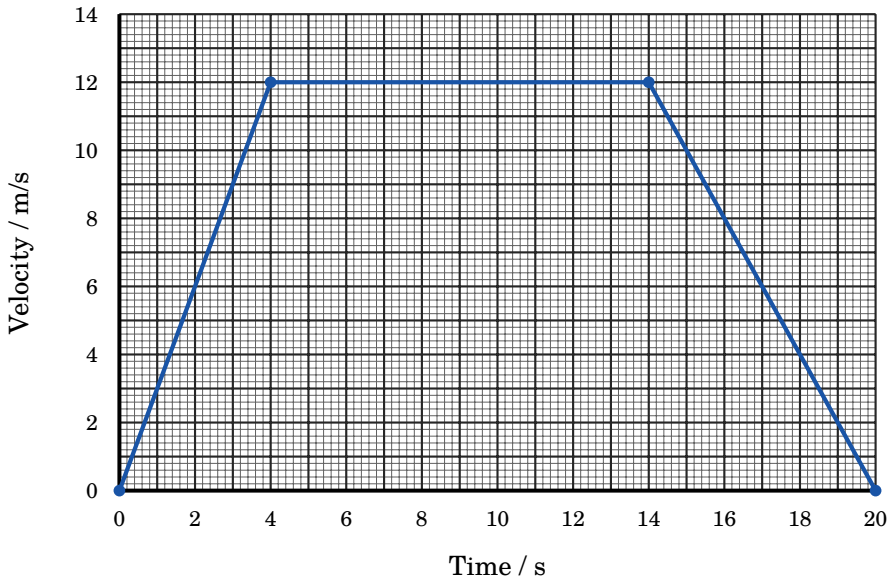
Section 1 — Gradient = acceleration

aP5-55

The gradient of a velocity–time graph is the acceleration. Same formula, one axis different:

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

Here the y values are **velocities** and the x values are **times**, so the gradient comes out as an **acceleration** in m/s^2. A flat line gives a gradient of zero: the velocity is not changing, so the acceleration is zero and the tram is still moving at a **constant velocity**.



Worked example — copy this shape

The first 4 seconds, as the tram pulls away. Read the two ends of that straight section off the graph:

point 1 = (0 s, 0 m/s) point 2 = (4 s, 12 m/s)

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{12 - 0}{4 - 0} = \frac{12}{4} = \mathbf{3\ m/s^2}$$

1. What is the tram's acceleration between **4 s and 14 s**? **0 m/s²**

Point 1 = (4, 12), point 2 = (14, 12), so gradient = (12 − 12) ÷ (14 − 4) = 0 ÷ 10 = 0. Zero acceleration means constant velocity — the tram is still travelling at 12 m/s the whole time.

2. What is the tram's acceleration between **14 s and 20 s**? Write down both points first. **−2 m/s²**

Point 1 = (14, 12), point 2 = (20, 0), so gradient = (0 − 12) ÷ (20 − 14) = −12 ÷ 6 = −2 m/s². The minus sign is the braking — keep it.

Check your two answers against the shape of the line. The tram takes 4 s to reach full speed and 6 s to stop, so the braking slope is the *gentler* of the two — its answer should come out smaller in size than the 3 m/s² you started with, and it does. The flat section should give exactly zero.

Section 2 — Area = distance

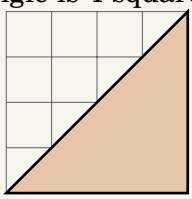
aP5-56, aP5-58, aP5-59

The area between the line and the time-axis is the distance travelled. Split a shape you do not recognise into ones you do — rectangles and triangles — work out each piece, then add them.

A second way to find the same area: count the squares. Count every square that is fully inside the shape. Then look at the squares the slanted edge cuts in half — each pair of half-squares, one on either side of the line, makes one more whole square. Add the two totals together.

Worked example — copy this shape

A small triangle, both ways. This triangle is 4 squares wide and 4 squares tall.



Formula: $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 4 \times 4 = \mathbf{8}$

Counting squares: 6 whole squares sit fully under the diagonal, and the 4 half-squares it cuts pair up into 2 more whole ones. $6 + 2 = \mathbf{8}$.

Both methods agree. Use the formula when the shape is a clean rectangle or triangle; count squares when it is not, or as a check.

3. The tram, 0 s to 4 s. This is a triangle: 4 s wide, 12 m/s tall.

Distance $= \frac{1}{2} \times 4 \times 12 =$ **24 m**

4. The tram, 4 s to 14 s. This is a rectangle: 10 s wide, 12 m/s tall.

Distance $= 10 \times 12 =$ **120 m**

5. The tram, 14 s to 20 s. A triangle again — but a wider one than the first. Read its base off the graph before you start.

Distance = **36 m**

Base 6 s, height 12 m/s, so $\frac{1}{2} \times 6 \times 12 = 36$ m. Wider base than the 0–4 s triangle, so a bigger distance — the tram covers more ground stopping than it did starting.

6. Total distance travelled, the whole 20 s journey. Add your three answers above.

Total distance = **180 m**

A quick scale check on the total. 180 m in 20 s at a top speed of 12 m/s is a sensible tram journey between two stops — not a sprint, not a crawl. If your total came out over a kilometre, or under ten metres, look for an arithmetic slip before moving on.

Section 3 — Average speed for non-uniform motion

aP5-43

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

Use the **total** distance — the one found from the area in Section 2 — divided by the **total** time on the graph. This is not the same as averaging the individual speeds you can read off the graph; it is the one number that describes the whole journey as if it had been travelled at a single, constant speed.

Worked example — copy this shape

The tram covers **180 m** in **20 s** in total (Section 2).

average speed $= 180 \div 20 =$ **9 m/s**

Check it against the graph. The tram's speed on this journey never drops below 0 m/s or rises above 12 m/s, so the average has to sit somewhere between those two numbers. **9 m/s does.** If your own average speed ever comes out higher than the fastest point on a graph, that is the signal to check the total distance or total time you used.

7. A different graph shows a cyclist covering **240 m** in **30 s**, made up of a speeding-up section, a steady middle, and a slowing-down section. What is their average speed for the whole 30 s?

8 m/s

$240 \div 30 = 8$ *m/s. You do not need the individual sections at all here — only the total distance and the total time.*

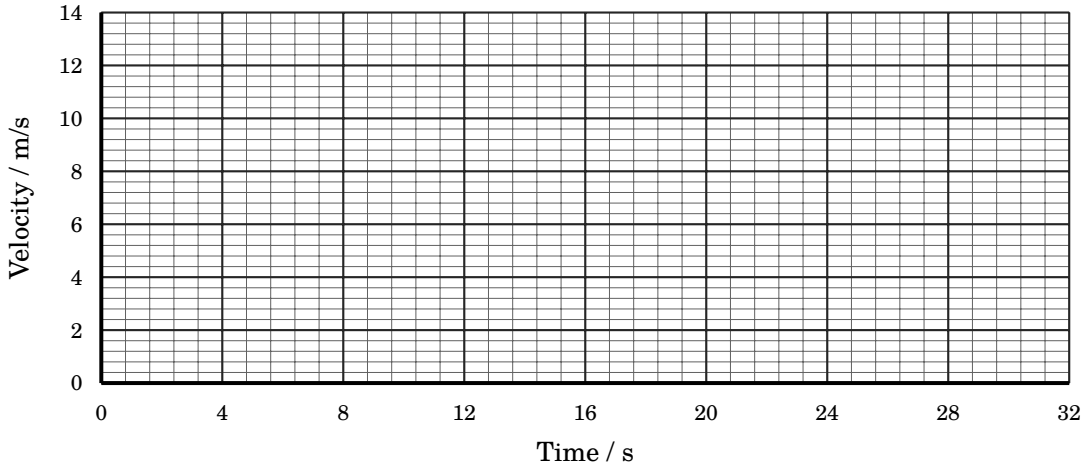
Section 4 — Your turn: the train

aP5-55, aP5-56, aP5-57, aP5-59, aP5-43

A train pulls out of a station, builds up speed in two stages, runs at line speed, then brakes into the next station. Its velocity was recorded every 4 seconds.

Time / s	0	4	8	12	16	20	24	28	32
Velocity / m/s	0	0	8	8	12	12	12	12	0

8. Plot the graph. Time on the horizontal axis, velocity on the vertical. Plot every point from the table, then join them up with a straight line between each pair — do not curve it. **Six straight sections** should appear.



9. What is the train's acceleration between **4 s and 8 s**? **2 m/s^2**

Point 1 $= (4, 0)$*, point 2* $= (8, 8)$*, so gradient* $= (8 - 0) \div (8 - 4) = 8 \div 4 = 2\ m/s^2$.

10. What is the train's acceleration between **12 s and 16 s**? **1 m/s^2**

Point 1 $= (12, 8)$*, point 2* $= (16, 12)$*, so gradient* $= (12 - 8) \div (16 - 12) = 4 \div 4 = 1\ m/s^2$*. The second stage of speeding up is gentler than the first.*

11. What is the train's acceleration between **28 s and 32 s**? **-3 m/s^2**

Point 1 $= (28, 12)$*, point 2* $= (32, 0)$*, so gradient* $= (0 - 12) \div (32 - 28) = -12 \div 4 = -3\ m/s^2$*. Braking is the sharpest change on the whole journey.*

12. Find the distance travelled in each section, using the area under your graph, and add them up. One section needs splitting into two shapes before you can do it.

Section	**Shape under the line**	**Distance / m**
0–4 s	**flat on the axis — no area**	**0**
4–8 s	**triangle**	$\frac{1}{2} \times 4 \times 8 = 16$
8–12 s	**rectangle**	$4 \times 8 = 32$
12–16 s	**rectangle + triangle**	$32 + 8 = 40$
16–28 s	**rectangle**	$12 \times 12 = 144$
28–32 s	**triangle**	$\frac{1}{2} \times 4 \times 12 = 24$
Total		**256**

The 12–16 s section is the one that needs splitting: a rectangle 4 s wide and 8 m/s tall sitting underneath ($4 \times 8 = 32$)*, with a triangle on top of it 4 s wide and 4 m/s tall* ($\frac{1}{2} \times 4 \times 4 = 8$)*.* $32 + 8 = 40$ *m.*

13. How far did the train travel between the two stations? **256 m**

14. Find the train's average speed for the whole 32 seconds. **8 m/s**

$256 \div 32 = 8\ m/s.$

Two checks on the total. The train's top speed was 12 m/s, so an average of 8 m/s sits sensibly below it — and the 16–28 s section alone is 144 m, more than half the journey, which fits: that is where the train is both quickest and running longest.